

Evaluation of lignin and cellulose nanoparticles from bacterial-deconstructed sugarcane bagasse for effective quercetin delivery

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Highlights

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Ectopseudomonas mendocina NITDDGIFE11 effectively breaks down sugarcane bagasse.

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Lignin and cellulose nanoparticles produced measured 279.4 nm and 207.25 nm.

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Encapsulation efficiency of LNP_I-Q and CNP_I-Q was 84.41 % and 62.64 %, respectively.

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Nanoparticles demonstrated sustained quercetin release over six days.

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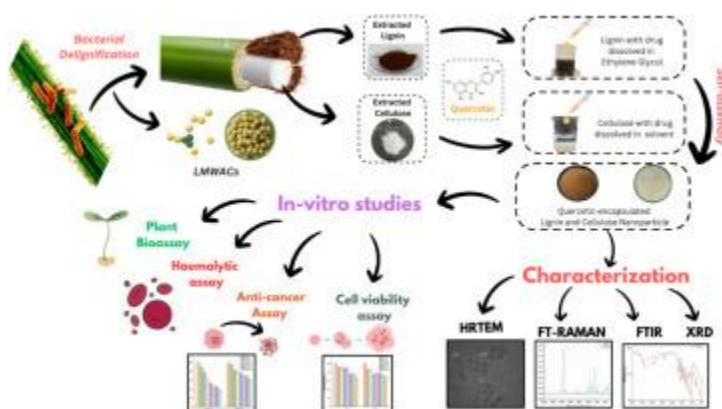
Nanoparticles enhanced growth in *Vigna radiata* with minimal hemolytic effects.

Abstract

This study attempts an innovative biorefinery strategy to valorize sugarcane bagasse (SCB), producing nanomaterials for targeted drug delivery for the hydrophobic drug Quercetin. Bacterial strain *Ectopseudomonas mendocina* NITDDGIFE11 was employed to delignify Sugarcane bagasse (SCB) *via* batch-submerged fermentation, and analyzed for alteration in its structural and chemical content. The extracted biopolymers lignin and cellulose, characterized *via* FESEM, XRD, and FTIR, were transformed into their respective lignin (LNP_I) and cellulose (CNP_I) nanoparticles, found to have size averaging 279.4 and 207.25 nm *via* DLS and validated by HRTEM (<150 nm), XRD, FTIR, and FT-RAMAN. Both

these nanoparticles were found to act as good delivery vehicles for Quercetin, where LNP_I-Q demonstrated superior encapsulation (84.413 %) and loading capacity (36.30 %) relative to CNP_I-Q and enhanced antioxidant activity compared to their free form. They exhibited sustained Quercetin release over 6 days and followed first-order kinetics. Reduced toxicity in HEK-293 cells revealed the host compatibility of the free LNP and CNP as a delivery system, while Quercetin-loaded forms exhibited significant anticancer activity, as examined in the triple-negative breast cancer cell line MDA-MB-231. Biocompatibility was further validated through haemolytic and phytotoxicity assays, confirming safety. Apart from biopolymer valorisation, a few interesting low molecular weight aromatic compounds (LMWACs) were also noticed in the bacterial-treated liquid fraction of SCB using LC-MS study. The outcome of this study not only provides a development of a drug delivery system from lignocellulosic biomass, but also may help in agricultural waste management and circular bioeconomy.

Graphical abstract



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Introduction

Sugarcane's remarkable efficiency in converting solar energy into biomass and sugar positions it as a key crop in tropical and subtropical climates [1,2]. Global sugarcane production from 2018 to 19 to 2022-23 reached 883.69 million metric tons, averaging 176.738 million metric tons annually [3]. A significant byproduct of this industry, sugarcane bagasse (SCB), presents both challenges and opportunities. SCB, a lignocellulosic material (LCB) composed of cellulose (35–45 %), hemicellulose (26–35 %), and lignin (11–25 %) [4], can serve as a valuable feedstock for electricity, bioethanol, and animal feed [[5], [6], [7]]. However, its utilization remains suboptimal, with substantial quantities being discarded or burned, which contributes to environmental concerns [1]. The inherent recalcitrance of

SCB is attributed to the intricate interactions between its constituents and the robust structure of lignin and polysaccharides, which impedes its efficient biological and biochemical degradation [8]. While traditional lignin extraction from sugarcane bagasse relies on organic solvents, recent efforts are increasing through the use of innovative techniques. Pre-treatment methods, also known as delignification, can be achieved through thermochemical methods and biological approaches employing enzymes or microbes, providing a more sustainable alternative. Previous reports state that conventional chemical extraction methods, which often rely on harsh and costly organic solvents, generate phenolic and other aromatic inhibitors and cause cellulose and hemicellulose degradation, which increases the carbohydrate loss and decreases the yield and quality of biofuel precursors [9,10]. Bacterial delignification stands out as an efficient and environmentally friendly pretreatment method compared to chemical and fungal approaches [9,11], as it can succeed and function in a wider range of pH, temperature, and oxygen levels, reducing processing time and costs, allowing efficient lignin breakdown and preserving valuable cellulose in various industrial and natural environments where fungal activity may be limited [10]. Ligninolytic enzymes produced by bacterial species are used in wood fibre bleaching as part of environmentally friendly preparatory procedures [12].

Pre-treatment techniques are essential for enhancing the accessibility of cellulose and hemicellulose while removing lignin. These methods aim to augment cellulose crystallinity, facilitate lignin removal, and improve the porosity and surface area of the biomass [13]. It is observed that the extracted biopolymers, cellulose, and lignin from LCB aided the preparation of functional substances such as membranes, aerogels, and microspheres, which had the encouraging potential to be used in the emerging fields of skin tissue engineering, carbon capture, and drug delivery [14,15]. Apart from improving pre-treatment strategies for biofuel research, development on using novel materials derived from sustainable bioresources, in the form of nano-celluloses, lignin nanoparticles, carbon dots, valuable chemicals, and biofuels, is gaining attention due to their diverse applications. These materials possess exceptional physical properties, making them suitable for various high-value products while minimizing their environmental impact [[14], [15], [16], [17], [18], [19]]. In this regard, extensive research has explored nanoparticles derived from both natural and synthetic polymers to enhance drug efficacy and bioavailability, but alarms with respect to the environmental impact and biocompatibility of synthetic polymers. These now impelled interest in the search for sustainable alternatives. Natural polymers, including gum acacia, liposomes, sodium alginate, chitosan, and gelatin, have been extensively investigated for their applications in drug delivery [20,21]. In this territory, plant dietary fibers, primarily composed of cellulose, hemicellulose, and lignin, present significant advantages due to their abundance, renewability,

biocompatibility, and potential for modification into functional nanomaterials [22,23]. In general, nanoparticles (NPs) improve their efficacy against Gram-positive and Gram-negative bacteria, as well as drug-resistant bacteria, when studied in combination with specific antibiotics and silver nanoparticles, zinc oxide nanoparticles, and Cu₂O-NP synergy [24,25]. Moreover, nanoparticles have been found to improve the intracellular delivery of hydrophobic drugs and improve the efficacy of specific therapeutics such as anticancer agents [26].

Among the various herbal anticancer agents, Quercetin (3',3'4',5,7-pentahydroxyflavone), a naturally occurring flavonoid that is abundant in various plant-based foods, including acai berries, onions, broccoli, apples, olive oil, and red wine, is typically found in glycosylated forms [27]. It has attracted the attention of dietitians and medicinal chemists due to its diverse array of pharmacological activities, including anticancer, antioxidant, anti-inflammatory, anti-pathogenic microorganism, and cardioprotective effects, with notably low cytotoxicity [28,29]. It is an effective antioxidant that has a well-documented role in decreasing different human cancers. On the other hand, the highly toxic effect of quercetin against cancer cells is accompanied by little or no side effects or harm to normal cells. It is observed that it exerts direct proapoptotic effects on tumor cells, facilitating the inhibitory effect in the progression of human cancers. The anticancer effect of quercetin has been documented in numerous *in vitro* and *in vivo* studies that involved several cell lines and animal models [30,31].

However, the clinical application of Quercetin is significantly hampered by its poor oral bioavailability, necessitating extensive efforts to enhance its delivery and absorption [32]. Several attempts have been made to develop delivery systems to improve quercetin bioavailability, such as polymeric nanoparticles, which have improved oral bioavailability and sustained release but are limited by reproducibility and toxicity [20,33,34]. Liposomes and solid lipid nanoparticles improve solubility and intestinal absorptivity, but their storage instability and poor solubility in aqueous gastric and intestinal fluids limit their usage [28,35]. Polymeric Micelles are easily formed yet unstable in gastrointestinal fluids, but improve solubilization and anticancer efficiency [36,37]. Among the various strategies for formulating nanoparticle-based drug delivery systems, [38] suggested a supplementary methodology to nanoparticle-based delivery systems by effectively associating biological effects with chemical identities, hence enabling the judicious selection of therapeutic candidates from intricate natural matrices. These methods offer considerable benefits by enhancing the solubility, stability, and bioavailability of natural substances, thereby optimizing their therapeutic effectiveness.

Our previous work with chitosan-based formulations [20] demonstrated enhanced drug delivery at higher intestinal pH levels, potentially improving the oral bioavailability of quercetin. However, improvement is still in demand as enhanced bioavailability of this drug in the near future is expected to bring this product to the head of therapeutic agents for the treatment of human disease. The usefulness of quercetin delivery systems is intensely influenced by two critical factors: the intrinsic properties of the materials used and the complex conditions of the gastrointestinal environment. The selection and design of materials for these delivery systems is thus of utmost importance, which remains underexplored. Among natural materials, lignocellulosic biomass (LCB) possesses a significant technical and economic impact. Cellulose and lignin are the biomass fractions that offer great interest due to their abundance, low cost, safety, and renewability. Lignin, as a byproduct from the pulping industry and biorefineries, is chiefly exploited as a fuel for energy production. Most of the research is focused on Kraft lignin [11]. It is interesting to explore other types of resources for the extraction of cellulose and lignin with environmentally friendly treatments. Bacterial treatment for the deconstruction of LCB has gained momentum nowadays [[39], [40], [41]]. Bacteria display promise in lignocellulolytic breakdown because of their innate ability to adapt and grow under both optimum and extreme conditions.

Apart from this, previous studies indicated that LCB delignification may be significantly influenced by the generation of biosurfactants. It is reported that some bacterial species from the *Pseudomonas* (*Pseudomonas aeruginosa*, *Pseudomonas fluorescens*) and *Bacillus* (*Bacillus safensis* J2) genus can produce biosurfactants like rhamnolipids using SCB, which facilitates lignin's breakdown and enhances enzymatic cellulose degradation for biofuel production [[42], [43], [44]]. Bacterial pretreatment enhances biomass accessibility by breaking down complex structures, facilitating the separation of its components [45]. The production of lignin and cellulose nanoparticles from sugarcane bagasse (SCB) holds promise as biodegradable nanomaterials for drug delivery, offering improved drug stability and controlled release [46]. However, synthesis faces many difficulties [22,47], and overcoming those is still challenging. From the above-mentioned previous studies, it could be seen that there is still demand for proper application of quercetin, as well as the exploitation of cellulose and lignin for the development of biopolymer-based drug delivery systems, which are still limited.

In this study, the potential of *Ectopseudomonas mendocina* NITDDGIFE11 (Accession no MH091331), which is an aerobic, rod-shaped, Gram-negative bacterium belonging to the family *Pseudomonadaceae*, was reported to produce biosurfactant [48] has been explored for the deconstruction of sugar cane bagasse (SCB) to transform both lignin and cellulose extracted into their respective nanoparticles for evaluating their role as drug delivery

vehicles for Quercetin, capitalizing on their inherent biocompatibility and tunable release properties. Furthermore, the comprehensive evaluation includes not only encapsulation efficiency and drug release kinetics but also detailed biocompatibility assays on human erythrocytes and *Vigna radiata*, providing a holistic assessment of the system's safety and efficacy. Second, the liquid fraction of treated SCB was examined for valuable low-molecular-weight aromatic compounds (LMWACs) using LC-MS, which can contribute to the overall biorefinery value. The outcome of this study may provide a sustainable bacterial treatment for designing biopolymer-based drug delivery systems to enhance the bioavailability of a hydrophobic Quercetin drug and produce valuable chemicals from lignocellulosic biomass, which may contribute to sustainability in agro-industrial waste management and the circular bioeconomy.

Reagents and materials

The bacterial strain used was *Ectopseudomonas mendocina* NITDDGIFE11 (NCBI GenBank accession no. MH091331), isolated in our lab from the effluent of the iron factory of Burdwan District, India. It can decolourise lignin mimicking dyes (supplementary Table 1), and has been explored for its potential to delignify LCB. *Vigna radiata* were procured from a local market in Durgapur, West Bengal, India. The Human embryonic kidney (HEK-293) and breast cancer cell lines (HEK-293, MDA-MB-231) were procured

Batch treatment of SCB with *E. mendocina* NITDDGIFE11: its growth profile

In recent times, biorefinery approaches are required for lignocellulosic industrial solid wastes that will not only emphasize on generating valuables from waste but also lead to complete waste utilization. Bacterial strains with ligninolytic and cellulolytic potential may be utilized, which can lead to the generation of cellulose or lignin-rich biomass, respectively. These compounds can be further utilized for the production of valuable products like cellulose nanocrystals or lignin

Conclusion

This study attempts to valorize lignin and cellulose, both extracted from the solid fraction of sugarcane bagasse after bacterial pretreatment. Both polymers were successfully converted into their respective nanoparticles (LNP and CNP), which were further investigated for their biomedical applications in the form of DDS to enhance the bioavailability of the hydrophobic drug, Quercetin. First, this strategy specifically shows sequential bacterial–alkali treatment of SCB for the extraction of

CRediT authorship contribution statement

Gaurav Singh: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Abhishek Verma:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Ankit Srivastava:** Visualization, Methodology, Investigation. **Subash Chandra Gupta:** Validation, Supervision, Data curation. **Tamal Mandal:** Writing – review & editing, Validation, Supervision. **Dalia Dasgupta Mandal:** Writing – review & editing,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors express their gratitude to the National Institute of Technology Durgapur, located in Durgapur, West Bengal, Bharat, as well as the Ministry of Education, Government of India, for their financial support. The authors are grateful to the Department of Central Scientific Services and the Indian Association for the Cultivation of Science (IACS) for providing access to the DLS and Zeta potential facilities. The author also conveys appreciation to the Centre of Excellence at the National